MILITARY SPECIFICATION

INVERTER, VIBRATOR PP-1703( )/U

1. SCOPE

* 1.1 This specification covers one type of portable vibrator-type power supply, for converting direct current to alternating current; designated Inverter, Vibrator PP-1703( )/U. (See 6.3)

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
QQ-S-781	Steel Strapping, Flat
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Communication Equipment

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MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electrical Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-DL-356910	Inverter, Vibrator PP-1703( )/U

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

* 3.1 Preproduction Samples. - The contractor shall furnish three (3) preproduction samples of Inverter, Vibrator PP-1703( )/U for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2(d)(1))

* 3.2 Construction. - The equipment shall be constructed in accordance with the requirements of this specification, and of Drawing SC-DL-356910 (including the requirements for parts, materials, and processes thereon). (See 4.14)

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* 3.3 Parts, materials, and processes. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 6.8)

3.4 Finish, protective. - Equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

* 3.5 Marking. - Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification. (See 4.4)

3.5.1 Visibility. - Whenever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Serial numbers. - Each vibrator inverter shall be serial numbered.

* 3.6 Input voltage. - The vibrator inverter shall be capable of operating, without failures of any of the parts, from an electrical system having a voltage range of from 20 vdc to 32 vdc.

* 3.7 Output voltage. - The equipment shall have a single phase output of 115 ± 5 vac, 60 ± 3 Hz when subjected to the test specified in 4.7.2.

3.8 Service conditions. - The equipment shall meet the following service conditions:

* 3.8.1 Operation (general). - The equipment shall operate under environmental service conditions specified in 3.8.2 and 3.8.3 with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.8. Approval of the preproduction sample (see 3.1) shall be considered as compliance with this requirement.

* 3.8.2 Service conditions operational requirements. - Unless otherwise specified, prior to, during, and after testing as required for each of the following service conditions, the equipment shall conform to the requirements of 3.7.

* 3.8.2.1 Low temperature. - See 4.8.1

* 3.8.2.2 High temperature. - See 4.8.2

* 3.8.2.3 Humidity (moisture-resistance). - See 4.8.3

* 3.8.2.4 Altitude (elevation). - See 4.8.4

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* 3.8.2.5 Bounce (vibration) and shock. - The equipment shall be capable of meeting the requirements of table I.

Table I Bounce and Shock

Para. Heading	Test Para.	Performance Rqmts.
Bounce (vibration)	4.8.5	Specified performance (See 3.8.2 and Note A). No physical damage except minor surface abrasion.
Shock; bench-handling	4.8.6	Specified performance (See 3.8.2) No physical damage.
Shock; drop	4.8.7	Operable. Any physical damage shall be minor only.

NOTE A: With the equipment shock-mounts (if any)blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz.

* 3.8.3 Leakage (immersion). - The equipment shall show no evidence of leakage after it is tested as specified in 4.8.8.

* 3.9 Electromagnetic interference characteristics. - The equipment shall meet the elctromagnetic interference requirements specified in MIL-STD-461, table A-II-MIL-I-11748, Equipment Class IIIc. Tests and test methods shall be as specified in MIL-STD-462. (See 4.9) If the contractor considers it desirable to modify the equipment proper to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

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3.10 Life. - The equipment shall be capable of operating for a period of 500 hours without any failure. The vibrator shall be capable of providing a minimum of 500 hours of trouble-free service when tested as specified in 4.10. Failure of the vibrator to start at the reduced input voltage shall constitute failure of the vibrator. At the end of the 500 hour life test, the output voltage shall be within 10 percent of the voltage obtained at the start of the test.

* 3.11 Overall treatment (moisture and fungus proofing - treatment (MFP)). - Moisture and fungus proofing treatment shall be in accordance with MIL-T-152. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.11. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened surfaces.

* 3.12 Final operation check. - When subjected to the test specified in 4.7.3, the equipment shall meet the requirements of 3.7.

* 3.13 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.12)

3.14 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5)

3.15 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on order. (See 6.2(g) and 6.2(h).

* 3.16 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

(1) General requirements for plastic material and plastic parts.

(2) Wiring and cabling, including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

(3) Flux and cleaning agents for solder.

(4) Soldering

(5) Cleaning

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(6) Riveting

(7) General requirements for securing of parts.

(8) Welding

(9) Tropicalization.

(10) Mechanical operation of controls.

(11) Self-tapping screws

(b) In this Specification:

(1) Marking

(2) Finish

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.15)

* 4.3 Preproduction inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B and environmental/design (See Tables III, IV and II respectively). The inspection requirements of 4.4, group A, and group B shall be performed on all samples prior to environmental/design inspection. Environmental/design inspection shall be performed on the samples specified and in the sequence shown in table II.

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Table II Environmental/design inspection

Inspection	Req. Para.	Insp. Para.
SAMPLE NO. 1		
Low temperature	3.8.2.1	4.8.1
High temperature	3.8.2.2	4.8.2
Humidity (moisture-resistance)	3.8.2.3	4.8.3
SAMPLE NO. 2		
Altitude (elevation)	3.8.2.4	4.8.4
Electromagnetic interference	3.9	4.9
Life	3.10	4.10
SAMPLE NO. 3		
Leakage (immersion)	3.8.3	4.8.8
Bounce (vibration)	3.8.2.5	4.8.5
Shock, bench handling	3.8.2.5	4.8.6
Shock, drop	3.8.2.5	4.8.7
Fungus	3.11	4.11

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidary documents as part of the inspection before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

* 4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to bounce preconditioning (see 3.14) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (Cont'd)

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4.5 (Cont'd) (See 6.5) Test Equipment for Government verification inspection shall be made available by the contractor.

* 4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.7.3) shall be last. (See 6.7)

Table III - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.16	4.13	1.5 DPHU *	6.5 DPHU*
Electrical Tests				
Output	3.7	4.7.2	1.5 DPHU	**
Operational Test	3.12	4.7.3	0.1%	**

* DPHU - DEFECTS PER HUNDRED UNITS

** ALL DEFECTS ARE CONSIDERED MAJOR

* 4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table IV and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

Table IV - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Leakage (Immersion)	3.8.3	4.8.8	4.0%	*
Interchangeability	3.13	4.12	6.5%	*

* All defects are considered major.

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* 4.5.2.1 Group B sampling plans. - The group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	INSPECTION LEVEL
4.0%	S-4
6.5%	S-4

* 4.5.2.2 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

* 4.5.3 Group C inspection. - This inspection shall consist of the tests listed in Table V and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

Table V Group C inspection

Inspection	Req. Para.	Insp. Para.
Group I		
Low temperature	3.8.2.1	4.8.1
High temperature	3.8.2.2	4.8.2
Humidity	3.8.2.3	4.8.3
Group II		
Altitude (elevation)	3.8.2.4	4.8.4
Bounce (vibration)	3.8.2.5	4.8.5
Shock, bench-handling	3.8.2.5	4.8.6
Shock, drop	3.8.2.5	4.8.7
Group III		
Life	3.10	4.10
Group IV		
Electromagnetic interference	3.9	4.9
Fungus	3.11	4.11
Group V		
Equipment Verification Review	3.2	4.14

4.5.3.1 Sampling for Group C inspection. -

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* 4.5.3.1.1 Group I and Group II Table V. - One sample shall be selected at random for each group, from each 200 equipments produced and subjected to each inspection specified, in the order shown. The first samples shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

* 4.5.3.1.2 Group III and Group IV - Table V. - One sample shall be selected at random for each group, from each 500 equipments produced and subjected to each inspection specified, in the order shown. The first samples shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Group V - Table V. - The Equipment Verification Review (EVR) shall be performed on the initial production lot. No subsequent inspection for EVR will be required. (See 4.14)

* 4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective. (See 6.6)

* 4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract or order, provided they are resubjected to and pass group A inspection after repair of allvisible damage and replacement of vibrator upon completion of life test.

* 4.6 Bounce preconditioning. - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (See 3.14 and 4.5)

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* 4.7 Electrical tests. -

* 4.7.1 Input voltage. - Inverter, Vibrator PP-1703( )/U shall be operated from a variable dc power source to determine compliance with the requirements of 3.6. Input test voltages shall be as specified in the applicable test paragraph.

* 4.7.2 Output voltage. - With the equipment connected to the power source specified in 4.7.1, the output voltage and frequency shall be measured to determine compliance with paragraph 3.7 when operated under the following conditions:

Input Voltage (VDC)	Output Adj. Sw. Position	Resistive Load (Ohms)
30	1	333
28	2	250
26	3	200
24	4	164

* 4.7.3 Operational. - The vibrator inverter shall be operated from a 24 vdc source, into a resistive load. The output voltage and frequency shall be measured to determine conformance to 3.12.

* 4.8 Service condition tests. - The following tests shall be performed on Inverter, Vibrator PP-1703( )/U.

* 4.8.1 Low temperature (See 3.8.2.1 and 6.8). - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be - 80°F for 6 hours and the operating temperature shall be stabilized at - 40°F. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.8.2.

* 4.8.2 High temperature (See 3.8.2.2 and 6.8). - The equipment shall be tested in accordance with Method 501, Procedure I, of MIL-STD-810. Exposure to the internal chamber temperature of 160°F shall not exceed 4 hours. The operating temperature shall be stabilized at 132°F. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.8.2.

* 4.8.3 Humidity (moisture-resistance) (See 3.8.2.3 and 6.8). - The equipment shall be tested in accordance with Method 507, Procedure II of MIL-STD-810. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.8.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If the equipment fails to meet specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

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* 4.8.4 Altitude (elevation) (See 3.8.2.4 and 6.8). - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810 except that the nonoperating chamber internal pressure shall be 279 mm of Hg (11 in. of Hg or 25,000 ft. above sea level). Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.8.2.

* 4.8.5 Bounce (vibration) test. (See 3.8.2.5). - The equipment shall be tested in accordance with Method 514, Procedure XI, of MIL-STD-810. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.8.2. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439.

* 4.8.6 Shock test, bench-handling (See 3.8.2.5). - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to testing, the equipment shall meet specification requirements for the measurements specified in 3.8.2.

* 4.8.7 Shock test: drop (See 3.8.2.5). - The equipment shall be tested in accordance with Method 516, Procedure II, of MIL-STD-810, except that the height of drop shall be 12 inches. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.8.2.

* 4.8.8 Leakage (immersion) test (See 3.8.3). - The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810.

* 4.9 Electromagnetic interference characteristics. - Conformance of the test sample in accordance with 3.9 shall be determined by the contractor or an approved testing laboratory. Testing shall be in accordance with MIL-STD-462. The test item shall be operated with an input of 24 vdc and with a resistive load of 164 ohms. The OUTPUT ADJ switch shall be in the 4-24 V position. Prior to testing, the contractor shall furnish the contracting officer, for approval, a test plan which shall include a block diagram of the test set-up and a list of the test equipment utilized. Upon approval by the contracting officer, and provided all other requirements of this specification are met, the approved test sample shall be used as a model for visual comparison of production units.

* 4.10 Life test (See 3.10). - The equipment shall be operated with a duty cycle of five minutes on and one minute off, for a period of 500 hours to determine compliance with paragraph 3.10. The OUTPUT ADJ switch shall be in the 4-24 V position. The input voltage shall be 24 vdc and the load 164 ohms resistance.

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* 4.10.1 Starting. - At least once during every 50 hours of the life test, the equipment shall be tested for starting with input voltages of 20 vdc and 32 vdc. At each input voltage, the POWER ON/OFF switch shall be operated five times.

* 4.11 Fungus test (See 3.11). - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

* 4.12 Interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

* 4.13 Visual and mechanical inspection. - The equipment shall be examined for the defects listed in MIL-STD-252. (See 3.2 and 3.16)

* 4.14 Equipment Verification Review. -

4.14.1 The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction (par. 3.2). The EVR shall consist of the following:

4.14.2 An audit to establish that "as-built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

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4.14.3 An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.14.4 The contractor shall compile an EVR report (s) containing the following information:

(a) Identification of unit, assembly, module, or part and details proving compliance with 4.14.2 and 4.14.3 above.

(b) Discrepancies noted.

(c) Corrective action taken.

4.14.5 The contractor shall make available to the Government copies of his EVR report(s) no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

(a) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(b) As required, responsible personnel from each functional department available for discussions in their respective areas.

(c) Adequate administrative support for the EVR.

4.15 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

* 4.16 Rough handling test (preparation for delivery). - When rough handling test in accordance with MIL-P-116 is required by contract (See 6.2(d)(2)(c), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Inspection	Insp. Par.	Req. Par.
Operation (final)	4.7.3	3.12

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* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified. (See 6.2(b))

5.1.1 Level A.

5.1.1.1 Cleaning. - Each Inverter, Vibrator PP-1703( )/U shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each Inverter, Vibrator PP-1703( )/U shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the method prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged Method I C-1.

5.1.1.4.2 Cable. - Starting at free end of cable coil in proportionate diamention, tie in three places withcotton twine conforming to T-T-871 and place within the compartment provided within the case.

5.1.1.4.3 Stowage. - Stow the spare fuses and vibrator, interrupter within the clips and compartment provided within the case. Close the cover of the case and secure all fastenings.

5.1.1.4.4 Inverter, Vibrator PP-1703( )/U. - Each Inverter, Vibrator PP-1703( )/U shall be packaged individually, in accordance with Method III of MIL-P-116. Cushion each Inverter, Vibrator PP-1703( )/U on all surfaces with pads or cells or both fabricated of fiberboard conforming to PPP-F-320, W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting corrugated fiberboard box conforming to PPP-B-636, W5c, class weather resistant. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid, closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. - Inverter, Vibrator PP-1703( )/U shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. - Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents.

5.2.1 Level A.

5.2.1.1 Palletized load. - A quantity of Inverter, Vibrator PP-1703( )/U, packaged as specified in 5.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "banded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproof containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2 or PPP-B-585 style 2 or 3, class 3. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. - A quantity of Inverter, Vibrator PP-1703( )/U, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Inverter, Vibrator PP-1703( )/U, packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. - A quantity of Inverter, Vibrator PP-1703( )/U, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

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5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Inverter, Vibrator PP-1703( )/U, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Inverter, Vibrator PP-1703( )/U is intended for use as a source of power for Tube Tester I-177( ) or Test Set, Electron Tube TV-7( )/U.

* 6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(c) The specific paragraphs of section 5 which are applicable to this procurement.

(d) Preproduction inspection:

(1) Three sample units of each item cited in section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.1)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon. (See 4.16)

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.15)

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(h) Quantity of tools and running spare parts required. (See 3.15)

(i) Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.11)

* 6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-1703M/U. The contractor should apply for nomenclature (Type Designation) in accordance with the applicable clause in the contract. (See 1.1)

* 6.4 Color. - A color chip to match the color specified (See 3.4) may be obtained, if required, upon request to: Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Ground, Maryland.

6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action. (See 4.5.3.2)

* 6.7 Location of operational inspection. - It is desirable that the operational inspection (4.7.3) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspection sample units) be sampled and inspected immediately prior to packaging. (See 4.5.1)

* 6.8 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, construction methods, etc.

* 6.8.1 Temperature. -

(a) Operating: Ambient temperature in the range of + 132°F to -40°F. (The high temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature not to exceed 6 hours, at any one time.

(b) Nonoperating: Exposure in the range of + 160°F to - 80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 6 hours, at any one time.

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* 6.8.2 Relative humidity. - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

* 6.8.3 Altitude (elevation). -

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 25,000 feet above sea level.

* 6.8.4 Orientation. - Storage in any position for a period of two years.

* 6.8.5 Immersion. - No leakage when placed in three feet of water for 2 hours.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-I-14603B(EL) Inverter, Vibrator PP-1703( )/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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